



r/LETfs • -5 meses

noletovictor

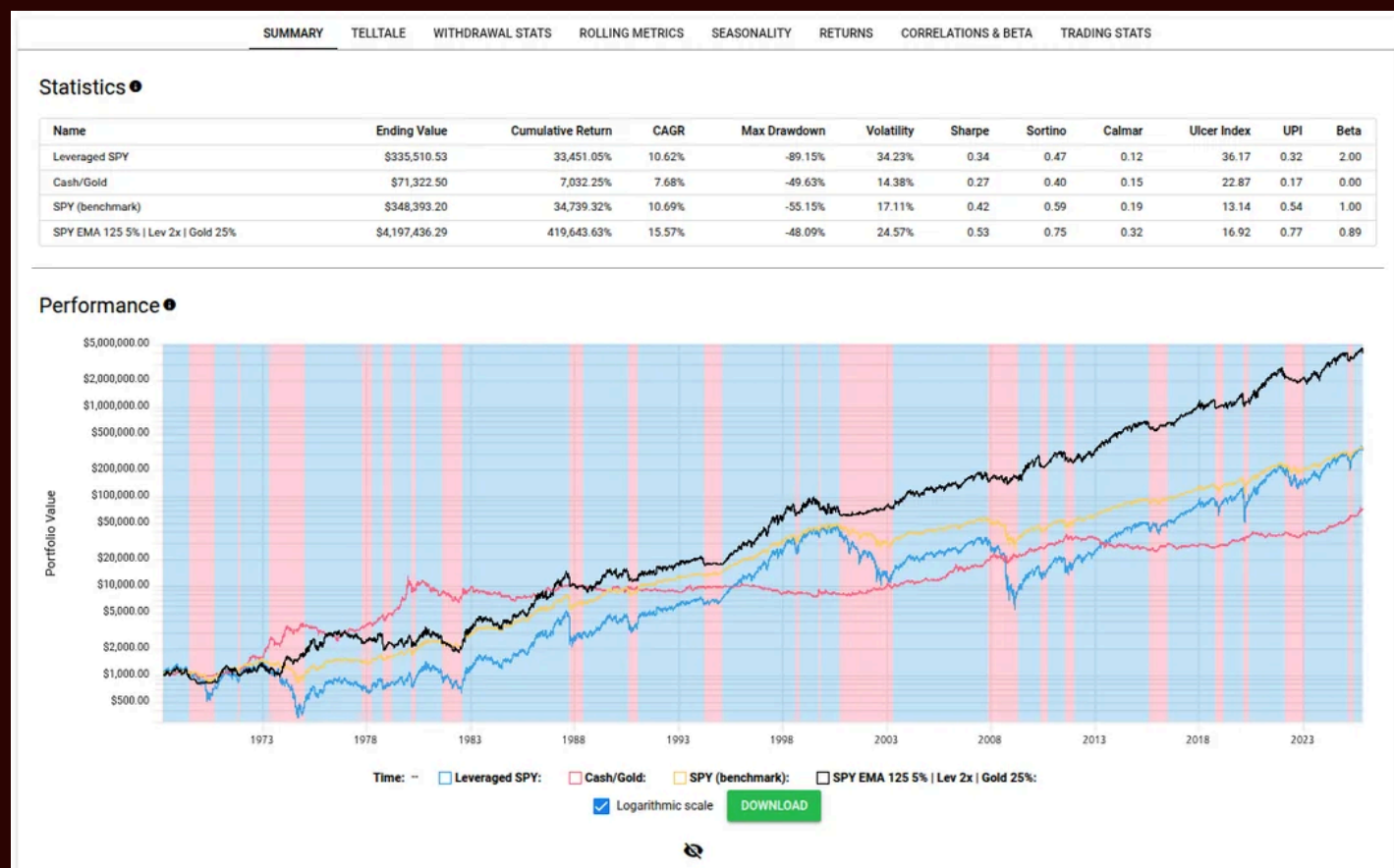
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The best optimization for Leveraged Rotation Strategy

TL;DR: In recent weeks I've been sharing the evolution of my study aimed at finding the best setup for a strategy that involves being invested in a leveraged ETF when the price of the underlying asset is above its moving average. In this post, I'm sharing the results obtained after analyzing **over 220,000 backtest results from 960 different combinations**.

The best result obtained was the SPY EMA 125 5% | Lev 2x | Gold 75% configuration. This setup achieved a **cumulative final result 12 times greater** than the buy and hold strategy on the SP500, **with a maximum drawdown 12.80% better**.

If you continue reading, I will explain the scoring algorithm process. At the end, I will also share other options/settings that are also relevant for those seeking higher returns (even if this comes at the cost of greater volatility and drawdowns).



SPY EMA 125 5% | Lev 2x | Gold 75%



Rank	Score	Sharpe	Sortino	Calmar	Max Drawdown	Win Rate
SPY (benchmark)	0	0.00	NaN%	0.00%	NaN%	NaN%
Total	44	0.76	77.27%	100.00%	15.57%	24.57%

Trading Stats

Briefly explaining my scoring algorithm, it consisted of comparing the backtest result with the benchmark (buy and hold of the underlying asset over the same period). The differences obtained (from all metrics, from all results) were averaged (within each time window), eliminating outliers (winsorization).

Using the time window averages for each setup, a score was calculated. There's no absolute rule or truth about how this should be done. However, I decided to use 3 metrics: **Calmar, Sharpe, and Sortino**.

- Calmar is the ratio between CAGR and the maximum drawdown.
- Sharpe penalizes volatility;
- Sortino penalizes negative volatility only;

The concept of these metrics (mainly sharpe and sortino) is quite interesting and worth further reading/study. However, I will not focus so much on this here.

The scoring for a time window was done using a weight of 0.5 for the average of the calmar ratio, 0.35 for the average of the sortino ratio, and 0.15 for the average of the sharpe ratio. The final score was obtained by taking a weighted average of the scores per time window (i.e., the scores from the 30-year backtests are more important than the scores from the 5-year backtests).

[At this link you can download the CSV file](#) (117 MB) containing over 220,000 backtest results. [At this link you can view the SQL query](#) that generated the strategy rankings.



1	SPY EMA 125 5% Lev 2x Gold 75%	0.9999999591776604
2	SPY EMA 125 5% Lev 2x Gold 50%	0.9999999548364659
3	SPY EMA 125 5% Lev 2x Gold 100%	0.9999999124301336
4	SPY EMA 125 5% Lev 2x Gold 25%	0.999999900785782
5	SPY SMA 150 3% Lev 2x Gold 100%	0.9999998403911121
6	SPY SMA 150 3% Lev 2x Gold 75%	0.9999996978065158
7	SPY EMA 125 5% Lev 3x Gold 75%	0.9999996024142688
8	SPY EMA 125 5% Lev 3x Gold 50%	0.9999995559531812
9	SPY EMA 125 5% Lev 2x Gold 0%	0.9999995190237191
10	SPY EMA 125 5% Lev 3x Gold 100%	0.9999993257580256

Top 10 results

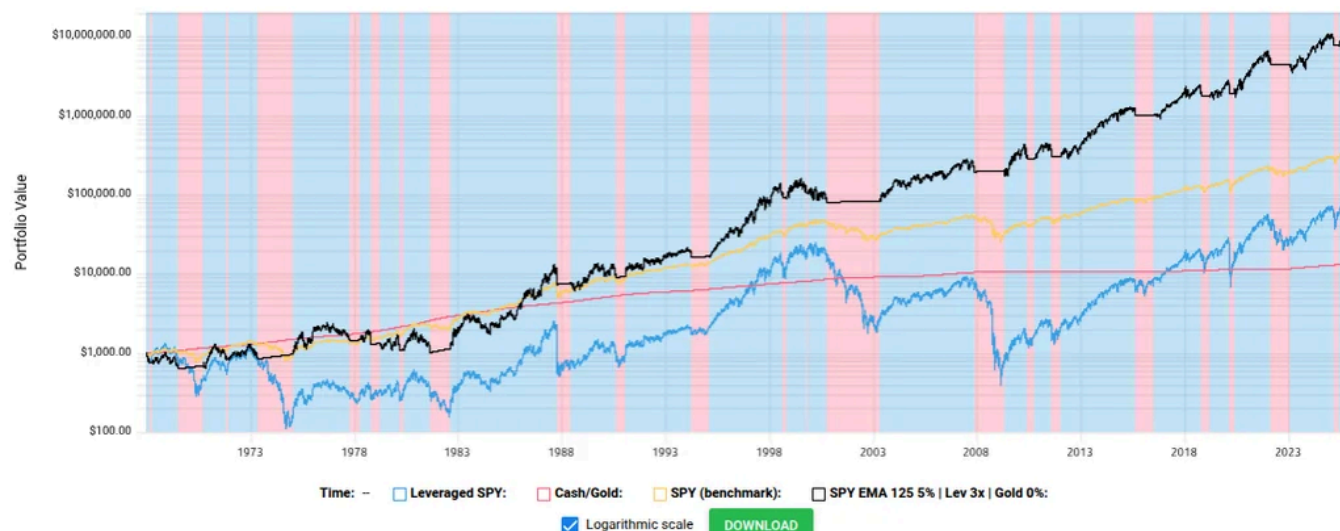
Based solely on this top 10 list, it's possible to draw some conclusions, such as: the EMA indicator generated better results than the SMA, and it's important to set a tolerance between 3% and 5%.

The 3x leverage appeared 3 times on this podium, practically at the end. **This is due to the calmar ratio.** This leverage does generate better results, but since this is accompanied by larger drawdowns, this metric is penalized.

However, since my goal is to use this strategy as part of my portfolio (and not entirely), I will proceed with the [SPY EMA 125 5% | Leverage 3x | Gold 0%](#).

Name	Ending Value	Cumulative Return	CAGR	Max Drawdown	Volatility	Sharpe	Sortino	Calmar	Ulcer Index	UPI	Beta
Leveraged SPY	\$76,460.36	7,546.04%	7.78%	-98.43%	51.29%	0.32	0.45	0.08	60.03	0.27	3.00
Cash/Gold	\$13,489.44	1,248.94%	4.60%	-0.00%	0.21%	NaN	NaN	12872.74	0.00	NaN	-0.00
SPY (benchmark)	\$338,334.29	33,733.43%	10.58%	-55.15%	17.09%	0.41	0.58	0.19	13.11	0.54	1.00
SPY EMA 125 5% Lev 3x Gold 0%	\$9,707,866.26	970,686.63%	17.19%	-57.88%	34.48%	0.50	0.71	0.30	25.86	0.67	1.35

Performance



SPY EMA 125 5% | Leverage 3x | Gold 0%

This strategy yielded a cumulative final result approximately 28 times greater, with a maximum drawdown virtually equal to that of buy and hold.

According to our ranking, we can obtain even better/higher values by allocating to gold during periods when the price is below the moving average; however, for practical reasons, I believe that:

1. It's easier to maintain the strategy using 0% or 100%;
2. It's more annoying having to deal with capital gains tax at both times;

Finally, if we compare it to the strategy that generated all this discussion, [SPY SMA 200 0% | Lev 3x | Gold 0%](#), we can see how these small adjustments completely changed the game.



Name	Ending Value	Cumulative Return	CAGR	Max Drawdown	Volatility	Sharpe	Sortino	Calmar	Ulcer Index	UPI	Beta
Leveraged SPY	\$103,811.96	10,281.20%	8.66%	-98.43%	51.87%	0.34	0.48	0.09	59.17	0.30	3.00
Cash	\$12,034.18	1,103.42%	4.55%	-0.00%	0.21%	NaN	NaN	12745.95	0.00	NaN	-0.00
SPY (benchmark)	\$328,624.58	32,762.46%	10.93%	-55.15%	17.29%	0.43	0.60	0.20	13.09	0.57	1.00
SPY SMA 200 0% Lev 3x Gold 0%	\$4,252,673.58	425,167.36%	16.13%	-71.51%	34.72%	0.48	0.67	0.23	32.73	0.51	1.34

Performance 📊



SPY SMA 200 0% | Lev 3x | Gold 0%

The important thing to note here is not only the difference in final result (whether CAGR or maximum drawdown, both of which were worse) **but also the trade statistics.**

Name	Switches	Switches per Year	Win Rate	% of Time	CAGR	Volatility
Leveraged SPY	21	0.38	80.95%	74.16%	24.02%	40.41%
Cash/Gold	21	0.38	100.00%	25.84%	4.63%	0.23%
SPY (benchmark)	0	0.00	NaN%	0.00%	NaN%	NaN%
Total	42	0.75	90.48%	100.00%	18.72%	34.83%

SPY EMA 125 5% | Lev 3x | Gold 0% — Trading Stats

Total trades: 42

Name	Switches	Switches per Year	Win Rate	% of Time	CAGR	Volatility
Leveraged SPY	161	2.88	21.12%	75.09%	20.05%	40.05%
Cash	161	2.88	100.00%	24.91%	4.97%	0.23%
SPY (benchmark)	0	0.00	NaN%	0.00%	NaN%	NaN%
Total	322	5.76	60.56%	100.00%	16.13%	34.72%

SPY SMA 200 0% | Lev 3x | Gold 0% — Trading Stats

Total trades: 322



Yes, the SMA 200 strategy achieved a higher final result than buy and hold. However, it was very interesting to discover how some small adjustments improved (and greatly improved) this result. **Not only did it improve the final result, but it also made it easier to maintain this strategy for decades.**

Conclusion

I believe I managed to say everything I wanted to. I tried to be as brief and direct as possible. I will be very happy to contribute to this discussion here and answer any questions about the methodology I used.

I am happy to make this small, but dedicated, contribution to the community. My goal is to continue with this strategy the next time the price crosses the moving average upwards. As I mentioned, I will dedicate about 25% of my capital to this.

I have heard some reports here of people investing 100% of their capital in leveraged ETFs, mainly 2x leveraged ones like SSO and QLD. I would (strongly) recommend in this case that they opt to use this 2x leveraged rotation strategy, as I mentioned at the beginning of the post.



45



86



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DysphoriaGML • há 5 meses

Great work, I wonder however if you took any precaution to avoid overfitting?

noletovictor **OP** • há 5 meses



DysphoriaGML • há 5 meses

You tested 960 combinations of strategies and you found the one/ones that perform better. However, statistically speaking the best ones could be better just by luck and the more you test, the more is likely you will find the right combination that will give you the best results, which is great. The problem arise once your strategy will face new unseen data, where your best strategy will break. This concept is called overfitting and it is true is finance as it is true in any other data-driven field of science. In other words, your strategy works great on the data you have at hand and you know already, but it does not generalise on new unseen data. So the target of your optimisation must be maximising the performance of your strategy on known data, but compare your strategy against other strategies based on their performance on unseen data. You can google cross-validation for more informations

Edit: typos and clarity



noletovictor **OP** • há 5 meses

In this case, I believe the study I conducted is reliable in this regard. For two reasons:

1. The 960 combinations were executed over approximately 230 different periods (variations of 5, 10, 15, 20, 25, and 30 years). Starting in 1970 and ending in 2025.
2. To obtain the final ranking, I can, if I want, create some filters. I did a test obtaining the final ranking only from backtests that went up to 2010 (imagine if I were doing the result in 2010). The ranking obtained was not very different from the current one. That is, if I had maintained one of the top positions in the ranking at that time, I would have obtained a good result.

No, absolutely no, study done with past data proves future performance success. However, I believe that what I did stressed the configurations in different time periods and with very significant crises.



Destrolas • há 5 meses • Editado há 5 meses

Your 2nd point is close to checking for overfitting but not quite there. If you were to take your 2010 ranking and compare to a post-2010 ranking only that would be a good overfitting check. By comparing 1970-2010 vs 1970-2025 you're already guaranteeing a correlation because most of the data set overlaps.

The easiest way to do this somewhat systematically would be to run a ranking stability/correlation between two parts of the data set. Basically:

- split the data in two pieces (where to split is mostly arbitrary)
- run methodology on both and generate separate rankings for the two data sets
- make a scatterplot where each point is a strategy and the X value is the ranking in data part 1 and the Y value is the ranking in data part 2



If it is overfit, it's possible that there are some parts with predictive power. You can try to isolate them. For example, it's possible that the ranking parameters have different levels of correlation. I'd expect max drawdown to be heavily dependent on the time period. If you do a ranking correlation using only max drawdown for ranking and there's no correlation, then you should remove it from the ranking system in favor of things with actual predictive power. You could do individual studies on the inputs as well like EMA length, portfolio composition, and so on to see if anything does have staying power.



DysphoriaGML • há 5 meses

Here, read some more so you can understand better what I mean:

<https://medium.com/@bauermartin101/pseudo-mathematics-and-financial-charlatanism-unmasking-the-dangers-of-backtest-overfitting-on-e84884bcb969>

Also, I am curious on how did you test all these combinations. Did you manually input everything on testfol.io?



noletovictor OP • há 5 meses

Thanks for the suggestion, I'll read it.

Regarding the combinations, I wrote a Python script to run the backtests using the testfol.io API.

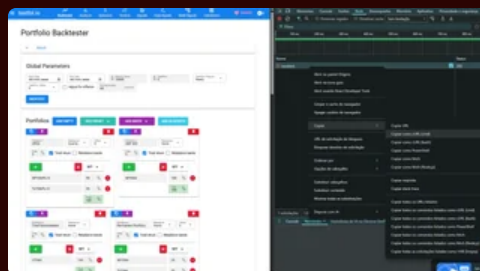


DysphoriaGML • há 5 meses

Mind sharing how to automate queries?



noletovictor OP • há 5 meses



Open your browser developer tools (in chrome, F12), go to the network tab and you'll see the endpoints. You can copy as curl. Once you understand the payload, you can modify and automate this on any backend language.



Thanks!



BAMred • há 5 meses

Do you have a link to the testfolio API?



noletovictor **OP** • há 5 meses

It's not a "public api". It's just the cURL requests they did. But it's public. Because the site itself is public.



seatosea_2020 • há 5 meses • Editado há 5 meses

I think generally here ppl talk about the SSO/ZROZ/GLD combo rather than pure 2x/3x

70/15/15

<https://testfol.io/tactical?s=huEuE5mfth8>

100

<https://testfol.io/tactical?s=0i8IhwBeCLp>



noletovictor **OP** • há 5 meses



It's a very interesting option, indeed. Having a maximum drawdown of only 36% over more than 50 years is impressive. In any case, I believe both options are interesting, and for those who want greater practicality, the LRS with EMA is a great fit.



Mycologist-Apart • há 5 meses

Do you use 5% above and below the ema to trigger the trade?



noletovictor **OP** • há 5 meses

**badjoeybad** • há 5 meses

curious to know if there is a particular reason for gold at 75% w/ cash instead of just going to 100% ? if you backtest starting 1/1/72 after nixon killed gold standard the returns are better with 100% gold, although slightly elevated max drawdown and ulcer index. but pretty minimal.

if you backtest to 1/1/00 which we can say is a starting point for the current era of fed manipulation/funny money, 100% gold does alot better with almost the same volatility and ulcer index. 2000gold below is 100% gold (ignore cash/gold header)

Results (25.88 years: 2000-01-03 - 2025-11-21)												
	BALANCE	TRADING	INFORMATIONAL	STATS	ROLLING	METRICS	SENSITIVITY	METRICS	CORRELATIONS & BETA	TRADING	STATS	
Statistics												
Name	Starting Value	Cumulative Return	CAGR	Max Drawdown	Volatility	Sharpe	Sortino	Calmar	Skew Index	UP	DOWN	Beta
Longest Gold	\$1,234.56	5352.0%	16.71%	-33.95%	24.38%	0.45	0.45	0.45	0.45	0.45	0.45	0.45
Gold/Gold	\$1,234.56	5352.0%	16.71%	-33.95%	24.38%	0.45	0.45	0.45	0.45	0.45	0.45	0.45
SPY (Benchmark)	\$1,234.56	450.0%	5.00%	-10.00%	15.00%	0.50	0.50	0.50	0.50	0.50	0.50	0.50
2000gold	\$1,234.56	5352.0%	16.71%	-33.95%	24.38%	0.45	0.45	0.45	0.45	0.45	0.45	0.45

Performance												
Strategy	Starting Value	Ending Value	Cumulative Return	CAGR	Max Drawdown	Volatility	Sharpe	Sortino	Calmar	Skew Index	UP	DOWN
Longest Gold	\$1,234.56	\$6,543.21	5352.0%	16.71%	-33.95%	24.38%	0.45	0.45	0.45	0.45	0.45	0.45

this is against \$54k end value, 5352% total, CAGR 16.71% max draw 33.95%, vol 24.38% and ulcer 12.61 for the proposed strategy.

seems like 100% gold is simpler and beneficial tweak to strategy. any particular reason for 25% cash?

**noletovictor** OP • há 5 meses

I forgot to mention that the period used for the backtests was from 1970 to 2025. It's interesting that we can test what the final results would be if we only considered, for example, backtests that started after 1990.

**badjoeybad** • há 5 meses

as i already noted above, the reason for the specific time period of post 2000 is due to the increasing manipulation of interest rates by the Fed. which is still ongoing. considering you seem to be getting defensive as opposed to actually thinking about the question i'll just go ahead and leave you to it bud.

**noletovictor** OP • há 5 meses

As I explained in the post, the final result obtained is based on an average of hundreds of backtests, with various different time windows. What you did was test only one time window. However large it may be (25 years), it is still only a specific period of time.

**Successful-Ad7038** • há 5 meses



badjoeybad • há 5 meses

again, this is a question that relates to a macro trend, a secular period of manipulated interest rates, MMT, greenspan put, helicopter ben, ZIRP and at the same time the two biggest gold buyers in the world, india and china, both see leaps in development and wealth.

so if you can't see the distinction then i have to say that i can explain it for you, but i cant understand it for you.



WorkSucks135 • há 5 meses

But you can't know how long that macro trend will continue, just like you couldn't know in 2000 that it would be the trend going forward. You are literally fitting to:

a secular period of manipulated interest rates, MMT, greenspan put, helicopter ben, ZIRP and at the same time the two biggest gold buyers in the world, india and china, both see leaps in development and wealth.

Even Wallstreet firms are no better than random at predicting future macro. The entire OP is an overfit too.



badjoeybad • há 5 meses

And you don't know that it will change.



u/CanadianDoc2019 • Patrocinado



Securitize just put a Nasdaq stock onchain in a real test of tokenized equities



ChemicalStats • há 5 meses

Could you please show us a some kind of block permutation analysis? Otherwise your risk of overfitted results is way to high for any meaningful discussion.



DysphoriaGML • há 5 meses

I've been reading some of your post (summaries from ChatGPT as I do not read German) How would you approach such overfitting problem? Block bootstrapping? Rolling CV?





variability for a strategy on a historical time series, thus being heavily based on assumptions of ergodicity. It's fine to describe how a strategy would have 'worked' but not to make it robust - certainly not, if you ran thousands and thousands of parameter combinations.

For statistical inferences, I'd opt for stationary block bootstrapping/permutating on a narrow set of parameters. If anything comes out statistically significant on a 0.001 level, I'd torture the winning parameter set with permutations in a markov chain monte carlo regime switching process. Of course, it's time consuming but whenever I see thousands of parameters and metric aggregation to scores, it usually breaks pretty quickly, hence, me asking for ergodicity-robust results.



DysphoriaGML • há 5 meses

I see, thank for the reply! Its pretty close to the pipeline I had in mind actually



ChemicalStats • há 5 meses

Glad it made sense! :) I think you'll see that having a more nuanced, more technical approach in place changes your view on what's optimal... :)



DysphoriaGML • há 5 meses

I come from a data background, ML and heavy stats. I need to get used to timeseries ;)



No_Loquat_183 • há 5 meses

how about with just qld?



BAMred • há 5 meses

OP's point is that QLD has a worse max drawdown and more volatility, so maybe it's not better.



noletovictor **OP** • há 5 meses



Ecstatic_Feeling7407 • há 5 meses

Thanks for your impressive work!

How about QQQ EMA 125 5% | Lev 3x?

I wonder if using QLD and TQQQ as leverage rotation strategy can yield better result than SPY



grimmjoww1983 • há 5 meses

Sorry for noob question but could you clarify if I am understanding your strategy correctly for spy ema 125 5% | leverage 3x | gold 0%. You are all in a 3x leverage etf like tqqq when spy price crosses above the 5% over 125 day ema ? And you exit 100% into cash when spy falls below it ? Great work with all the backtesting



noletovictor OP • há 5 meses

Exactly. We can use UPRO or SPXL as 3x SP500.



grimmjoww1983 • há 5 meses

Oh my bad.... Tqqq is 3x qqq... thank you



grimmjoww1983 • há 5 meses



You might have to wait a bit to buy in though



Is this correct?

Scenario A

Qqq EMA 125+ 5% > 100% tqqq.

Qqq Ema 125 -5% > 100% cash?

Scenario B

QqQ Ema 125 + 5% > 100% Tqqq.

Anything below this > 100% cash

Which scenario is correct? Is that what tolerance means?



noletovictor OP • há 5 meses

Tolerance is applied to the moving average value. If the moving average value is 100, with a 5% tolerance we have:

1. If the SPY price is greater than 105 (100+5%), we maintain the leveraged position;
2. If the SPY price is less than 95 (100-5%), we sell the leveraged position and keep the cash/gold value. And we wait until the price crosses again (from below).



Ecstatic_Feeling7407 • há 5 meses

Sorry for another noob question

I have been reading all of your studies and posts for the past few weeks

Do you mean after SPY price is less than 95 (100-5%), sell all leveraged position, then buy leverage etf again when the price is more than 95% or 105% of ema?

Once again thank you for your dedication and hard work!



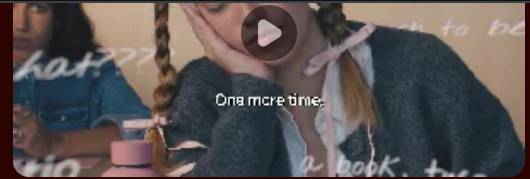
noletovictor OP • há 5 meses

No problem!! You buy again after 105%



u/Preply • Patrocinado



[Saber Mais](#)[preply.com](#)**bappyhappy18** • há 5 meses

This is incredibly insightful. thank you for putting in the work for this!

**noletovictor** **OP** • há 5 meses

Thank you very much!

**PecanPlan** • há 5 meses

48% is still a huge max drawdown to sit through. Most people will quit the system and miss the recovery.

Moving average market timing strategies work until they don't. I'd be careful with them. In my experience, seasonality strategies work better. But not always.

I do like the fact that your Cash/Gold piece is uncorrelated to your SPX piece. If you can find a group of ETFs that should grow and show low correlation and/or anticorrelation to one another, you can build a very low vol / low drawdown strategy that you can then leverage up with margin to your risk tolerance level.

**noletovictor** **OP** • há 5 meses

48% isn't high if you consider that the S&P 500 fell 55% in the same period. Once we have a benchmark, in my opinion, all numbers need to be analyzed relatively, not absolutely.

Regarding the group of ETFs with low or negative correlation, it is indeed worthwhile to study and look for something of that nature.

**PecanPlan** • há 5 meses

While your theoretical 48% is better than the actual 55% that occurred, both suck to sit through. Buffett did it twice but most won't have the stomach for it. Especially anyone older with real assets. You have to make 100% on the other side to wipe out a 50% loss and get to break even. That can take years.

If it gets worse than 50%, you are really screwed. At a 67% drawdown, you have to make 200% on the other side to break even. At 75%, you need 300% to break even.



Here's the problem. Let's say you hold on and the drawdown hits 48%. How do you know it's not going to 67%? 75%? Every percent worse it makes the road to recovery far longer. That's real pressure, and unless you've done it before I would suggest not brushing it off.



noletovictor OP • há 5 meses

I don't know. That's why I specified that I will only enter this strategy with 25% of my capital. If, in the worst-case scenario, an apocalyptic scenario where I lose everything, that's fine. But that hasn't even come close to happening with the tests from 1970 until now.

If all this means nothing (and I mean NOTHING at all) for the future, then why invest in anything?



prettycode • há 5 meses

It's not hard to stick to if it's performing on-par with or better than the benchmark. Being down 40% if the market is down 50% is only going to strengthen conviction.



confettofetti • há 5 meses

Could you elaborate on the sort of seasonality strategies you're referring to for using with leverages ETFs? Any good resources you would recommend to someone wanting to learn more?



PecanPlan • há 5 meses

See the StockTraders Almanac for the "Best 6 Months" cycle and the "4-year Presidential Cycle"

Basically you are in stocks in the cooler months, and in bonds in the warmer months. Except when an incumbent president is running for re-election. Then you are in stocks all year.



SiegfriedSigurd • há 5 meses

If you can find a group of ETFs that should grow and show low correlation and/or anticorrelation to one another, you can build a very low vol / low drawdown strategy that you can then leverage up with margin to your risk tolerance level.

Some other commenters have mentioned these before: VIX and inverse ETFs.

But the idea was dismissed. Maybe it's worth testing if some specific allocation of these can provide a greater option than gold in the "below SMA" times of the rotation.



here's your backtest with 50% GLD / 50% ZROZ instead : <https://testfol.io/tactical?s=1tKcckFam1X> (+2% CAGR)

Edit : same but for your second backtest : <https://testfol.io/tactical?s=6DJNqfEx9RZ> (+3% CAGR)

Overall i think your strategy (150 EMA 5% tolerance) is overfit like many in this sub.



noletovictor OP • há 5 meses

If it's overfit, how do you explain the fact that I limited the data to 2000 and 2010, and in those two cases obtained results/configurations that performed well until 2025?



Successful-Ad7038 • há 5 meses

What does "well" means ?



noletovictor OP • há 5 meses

Better cagr and max. drawdown compared to the underlying asset.



Successful-Ad7038 • há 5 meses

Almost all MA strategy are better than the underlying. What i wanted to say is that it's not by testing all scenarios with datas from the past that you'll find the best strategy even though it's a good start.

By the way does your 5% tolerance is on the crossing of the MA on the upside AND downside or only upside ?



noletovictor OP • há 5 meses

Upside and downside.



James__G • há 5 meses

Thanks for sharing this, it's very interesting analysis.

In terms of mitigating the overfitting, another thing you could do would be to re-run the analysis (pehaps with a smaller range of strats for time) with different nation's equity markets (esp the big ones, Japan, UK,



confettofetti • há 5 meses

I'm curious what people very concerned about over fitting think is a better option? From what I understand any moving average period is going to be somewhat arbitrary - an informed best guess? So I'd be interested to hear from people if they think this is too big of a flaw in all moving average strategies vs either the diversification with gold and bonds strategy or something else.



WorkSucks135 • há 5 meses

The "remedy" to overfitting is just to accept you don't need to be precise with the variables. Your future results will likely fall somewhere between the extremes of similarly constructed portfolios.



confettofetti • há 5 meses

Thanks for replying. Yeah I completely agree. I guess my question is why getting your best guess of what to go with from backtests like this wouldn't be considered reasonable, since you have to go with something? Is there something about the figures being suggested here?



blue_horse_shoe • há 5 meses

Not only were an absurdly large number of trades made, but they were of very poor quality, resulting in a very low win rate of 21%.

Excellent insight.



amc • há 5 meses • Editado há 5 meses

Finally, if we compare it to the strategy that generated all this discussion, [SPY SMA 200 0% | Lev 3x | Gold 0%](#), we can see how these small adjustments completely changed the game.

The important thing to note here is not only the difference in final result (whether CAGR or maximum drawdown, both of which were worse) **but also the trade statistics.**

Note this is not a fair comparison, as the SMA200 had no tolerance added which explains the trade stats.



- [SPY SMA 200 4% | Lev 3x](#)
- [SPY EMA 125 5% | Lev 3x](#)

Considering the 200 is way more followed/documented does this not appeal more to you than the 125?

However just a minor tweak on tolerance can highly affect metrics so we're likely into overfit territory.



noletovictor OP • há 5 meses

Indeed. But my goal wasn't to be "fair." Rather, it was to show the power of tolerance in avoiding unnecessary trades (false alerts), avoiding unnecessary fees, and maximizing returns. Because you can be sure that many people who know the 200-day SMA strategy would use 0% as tolerance simply because they don't know/disregard the fact that it uses a risk band.

Regarding the similar results, that's true. You can see that some strategies using SMA appear in the top 10. The final results of these "winning strategies" aren't that different.

I believe that because the 200-day SMA strategy is the most well-known/popular, it can (regardless of the magnitude) hinder you in terms of "having a lot of trading volume" when this happens. Imagine many (many indeed) people trying to buy/sell the asset "at the same time." This will either rapidly increase the asset price (at the time of purchase, making you pay more) or decrease the asset price (at the time of sale, making you sell your shares cheaper).

Therefore, regardless of whether you use SMA/EMA, I would opt for a "less popular" but equally effective timeframe.



Some-Afternoon-3642 • há 5 meses

Hi , have you considered and testing 1 additional "sell" criteria to further prevent drawdowns ? For example stop loss trail with x percent ?



noletovictor OP • há 5 meses

I thought about that. But I would need to develop the entire backtesting process because it's not possible to test this in the testfolio.



Some-Afternoon-3642 • há 5 meses

I'm currently building this setup in excel for testing



So if I understand it correctly, you decided to go with below, which has a CAGR= 17.19% and MDD of 57.88%?

However, since my goal is to use this strategy as part of my portfolio (and not entirely), I will proceed with the [SPY EMA 125 5% | Leverage 3x | Gold 0%](#).

There is a strategy using credit spreads as a signal, is it possible to test that on testfolio?

https://www.reddit.com/r/TradingEdge/comments/1nicant/i_developed_a_proprietary_strategy_trading_spxl/



confettofetti • há 5 meses • Editado há 5 meses

Thank you for taking the time to post this. I'm wondering if you've looked at, or have a sense of from seeing the data, how much the optimal ema/sma window changes for different timeframes?

I'm currently finding a moving average strategy more appealing than diversifying with bonds and gold. But I think there is a risk that if the first few trades weren't good ones I would question whether if this time is different and it doesn't work anymore. Are there any periods of time in the backtests where the optimum strategy actually becomes a bad one and a very different length of ema would have been necessary for success? Having said that your data, especially the success rates of your trades, are really promising and indicate this isn't the case unless the unsuccessful trades tend to be very unsuccessful and clustered together?

Edit: Just saw you explained in a comment that you have actually already done the below. I'm curious how much the results changed if you remember?

You may already know this but, along the same lines, one way of testing for the over fitting that people are asking about is to hold back some of your data so that you fit the model on say all but 20 years of data. And then you test your strategy on the remaining 20 years (which to your model is "the future") and see how well it performs to other strategies. You would do this for various combinations of years. And if the strategies that were good for the fitting data are also good for the testing data then you can see that your model is likely not over fit. But if there are instances where the best fitting strategies do quite badly on the test data this indicates a risk that this is likely to also happen in the future. For the in between results where the strategies are just mediocre for the testing data you would probably just say well I've got to pick something, this is as good as any. (Afraid it's been a while since I've done this with machine learning so I'm probably forgetting the correct technical terms - hope it's still useful info!)



Boys4Ever • há 5 meses

Problem with these strategies being past cannot contain the unknown future. Futile trying to formulate strategies where algorithms already anticipate retailer reaction to indicators such as MA.



Imswans • há 5 meses

what indicators do you prefer? in other words, when can you conclude the market is trending?



Boys4Ever • há 5 meses

Heikin-Ashi candles. Never lets me down.



Imswans • há 5 meses

thanks, i'll look into



dimonoid123 • há 5 meses

What about after tax returns after say 20 years vs buy and hold snp500 in a taxable account? Also account for bid-ask spread and commissions.

Even if not overfit, taxes will kill you.



Timely-Designer-2372 • há 5 meses

I'm not sure if I understand the strategy correctly:

If Spy is above 125 EMA, you go 100% SPY x2

If Spy is below 125 EMA, you go 75% Gold and 25% cash?

(Switch if the EMA 125 crossed by more than 5%)

If that's correct, I have 3 questions:

1. Why don't you use short term treasury instead of cash?
2. If you use data until 2023, does the strategy still perform great? Gold exploded last year
3. Have you tried HFEA and similar approaches with rebalancing to remain a specific ration between leverage etf and hedge?



noletovictor OP • há 5 meses



Regarding your questions:

1. It's entirely possible to use short-term treasury. In the case of "cash," it's simply a way to represent "a safe and easily liquid allocation."
2. I limited the records to those up to December 31, 2023, and the results obtained were no different. You can see them in the attachment below.
3. I haven't tried any strategy similar to HFEA (which holds a position with a fixed proportion). However, I don't rule it out as interesting. I think that in this sense of "strategy using leveraged assets," there will never be a single absolute solution. It's interesting to analyze each strategy and see which one best suits your style/profile.

The screenshot shows a SQL query in a query editor and its results in a table. The query filters for various financial metrics and leveraged assets. The results table has two columns: 'name' (text) and 'final_score' (double precision). It lists 10 different leveraged asset strategies with their corresponding final scores.

name	final_score
1 SPY EMA 125 5% Lev 2x Gold 25%	0.999866270763377
2 SPY EMA 125 5% Lev 2x Gold 0%	0.999975795241564
3 SPY EMA 125 5% Lev 2x Gold 50%	0.999609107238031
4 SPY SMA 150 3% Lev 2x Gold 25%	0.9996984072589463
5 SPY SMA 150 3% Lev 2x Gold 50%	0.9996631219938172
6 SPY EMA 125 5% Lev 2x Gold 75%	0.9996127470760309
7 SPY SMA 200 4% Lev 2x Gold 25%	0.9991811111111111
8 SPY SMA 150 4% Lev 2x Gold 25%	0.9991523645861402
9 SPY SMA 200 4% Lev 2x Gold 50%	0.9990411049140658
10 SPY SMA 150 3% Lev 2x Gold 75%	0.9988609226020639

Total rows: 960 Query complete 00:00:04.157

 **Timely-Designer-2372** • há 5 meses

Thank you for your response.

Would be interesting if you could also analyse those kind of HFEA approaches.

My plan is to do 2 different leverage etf approaches:

1. 3x TQQQ with SMA 200 (I already started).
2. 60% 3x UPRO, 20% TMF, 20% Gold with rebalance if UPRO is more than 70% or less than 50%.

2nd strategy has 180% S&P 500, so almost x2, but compared to x2 SMA/EMA you don't miss parts of the upswing after a drawdown. My only concern is the exact hedge position atm. TMF is said to be a good hedge, but it's performance is a catastrophe. That's why I think about shortterm treasury + Gold





Also, how is it possible that the 200 SMA strategy has more trades than the 125 EMA?



No_Building_659 • há 4 meses

Thank you for this! Would you be able to help me understand, do total returns include the borrowing costs because the borrowing costs are embedded into the returns of the fund?

I'm considering running the 2x version of this LRS as you have recommended but am still undecided on gold allocation. . Additionally, in the traditional /ZROZ/GLD portfolio, is SSO supposed to be sold when it crosses under as well or is that a buy/hold and rebalance portfolio?



undefined310 • há 4 meses

Great stuff! have you considered adding confirmation days to wait before transitioning between ETF and Cash?



Luckysmile19 • há 2 meses

Hello!

Thank you for your wonderful post!

I did the same for QQQ, QQQ SMA 250 | Lev 2x | Gold 100%. <https://testfol.io/tactical>
Tha Calmar ratio is even better but the drawdown is max 68% since 1995.



Mitchell11674 • há 5 meses



Snoo72726 • há 5 meses

Your post is fun and the effort is appreciated . but let's be clear:

What you built is a backtest curiosity, not a reliable allocation engine. It works because the dataset (SPY >2009) is easy not because the model is good.

A single-factor EMA crossover on a trending index tells you almost nothing about robustness. Trend-only LETF rotation is a fair-weather strategy that collapses the moment the market stops trending or volatility spikes.

If someone follows this without real macro + volatility conditioning, they're going to learn the hard way what professional quants already know:



Criar



confettofetti • há 5 meses

Are you suggesting that volatility (or something else) should also be an input to moving average strategies, or that there is nothing that can be done to compensate for this flaw in them?

What strategies do you prefer?